

SUMMARY

USC M.S. in Computer Science student and Machine Learning Engineer with an interdisciplinary background in machine learning and life sciences. Experienced in developing scalable ML platforms, production-oriented model serving systems, and reproducible PyTorch pipelines across multimodal biomedical learning, multi-omics modeling, protein design, and large language model research. Interested in building robust, interpretable, and deployable machine learning methods that bridge methodological research with scientific and therapeutic discovery.

EDUCATION

University of Southern California <i>M.S. in Computer Science</i>	Aug 2025 – May 2027 Los Angeles, California
Sogang University <i>B.S. in Computer Science and Engineering & Life Science (CGPA 3.81/4.3, Magna Cum Laude)</i>	Mar 2017 – Feb 2024 Seoul, Korea

SKILLS

Languages: Python, C/C++, Java, SQL, HCL, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: PyTorch, FastAPI, Pandas, NumPy, AlphaFold2, RDKit, Hugging Face, scikit-learn, Matplotlib
Technologies: AWS, Terraform, Docker, Databricks, ML/DL, RL, GPU/HPC, LLMs, Gen AI, RAG, Git/GitHub, GitLab CI/CD, Linux/Unix

PROFESSIONAL EXPERIENCE

Amgen <i>Machine Learning Engineer Grad Intern</i>	May 2026 – Present Thousand Oaks, California
- Engineered Python and FastAPI backend services for an internal machine learning platform, implementing durable asynchronous request processing while preserving real-time model inference for scientific and drug discovery workflows. - Built and maintained AWS infrastructure with Terraform and GitLab CI/CD to support scalable model deployment, workload isolation, and reliable production inference.	
University of Southern California <i>Graduate Student Researcher</i>	Aug 2025 – Present Los Angeles, California
- Designed a controlled 8-condition bias study (position/style/source) across 7 LLMs, revealing accuracy drops up to 51pp and exposing model-specific breakdowns invisible in single-factor tests. - Developed computational pipelines for AI-driven drug discovery, including a TCR-guided de novo binder design workflow using ProteinMPNN and AlphaFold2, and a reproducible Chemprop/RDKit framework for virtual screening and hERG cardiotoxicity prediction. - Built an LLM-based job recommender system that de-confounds candidate preference vs. employer qualification via dual-perspective modeling, and applied constraint-aware ranking with controllable policy alignment using LoRA fine-tuning.	
Bioinformatics & Machine Learning Lab , Chungbuk National University <i>Research Intern</i>	Aug 2024 – Jun 2025 Cheongju, Korea
- Developed a MAML-based few-shot meta-learning framework in PyTorch for predictive modeling on multi-omics datasets, improving cross-domain generalization in small-data settings. - Built a reproducible preprocessing pipeline for TCGA and GEO with Reactome-based pathway modeling and explainable AI, achieving a mean PR-AUC of 0.94 in the 5-shot setting on non-cancer cohorts and identifying candidate biomarker pathways through statistical validation.	
LikeLion , LikeLion Sogang <i>Full-stack Developer</i>	Mar 2021 – Dec 2021 Seoul, Korea
- Architected a full-stack URL repository service, developing a responsive and scalable UI with React and building RESTful APIs using Django REST Framework for core services and FastAPI for high-performance endpoints. - Implemented core features including URL saving, sharing, and a dynamic keyword search to streamline information retrieval and enhance user collaboration, driving a 25% increase in user engagement and monthly active collaborations.	

PUBLICATIONS

Journal / Conference Publications	
Kan, B.*, Yang, W.*, Nguyen, E., Yi, G., Yi, B., Yu, C., & Liu, Y.# De-conflating Preference and Qualification: Constrained Dual-Perspective Reasoning for Job Recommendation with Large Language Models. <i>arXiv preprint arXiv:2602.03097</i>	2026
Yi, G.*, Ju, J.*, & Jo, K.# Few-Shot Learning for Multi-Omics Disease Classification with a MAML-Based Model. <i>KIPS Transactions on Software and Data Engineering (KTSDE)</i>	2025
* indicates equal contribution, # indicates corresponding author.	
Oral Presentations	
Few-Shot Learning for Multi-Omics Disease Classification with a MAML-Based Model. <i>9th International Conference on Medical and Health Informatics (ICMHI)</i> , Kyoto, Japan	2025

TEACHING

9307261 Current Topics on Machine Learning , Chungbuk National University <i>Teaching Assistant</i> (with Prof. Kyuri Jo)	Spring 2025
5110134 Artificial Intelligence , Chungbuk National University <i>Teaching Assistant</i> (with Prof. Kyuri Jo)	Fall 2024

HONORS & AWARDS

Outstanding Startup Idea Award , Sogang University	Fall 2021
1st Place , LikeLion Hackathon	Fall 2021
1st Place , LikeLion Ideathon	Summer 2021

STANDARDIZED TEST SCORES

iBT TOEFL : 111 (Reading: 30/30, Listening: 29/30, Speaking: 28/30, Writing: 24/30)
GRE General Test : Verbal 161/170 (86%), Quant 170/170 (92%), Analytical Writing 4.5/6.0 (83%)